0

convert the drive speed reference from scaled integer units (0.01 hz) into engineering units (hz) for use in the simulation ramp calculation

MOVE
Source VFD_Reference
0
Dest Ref_Hz
0.0

DIV
Source A Ref_Hz
0.0
Source B 100
Dest Ref_Hz
0.0

1

drive run request based on start command, permission, and fault status

VFD_LogicCommand.1 Permissive_Runok Sim_DriverFault Sim_CommFault Drive_RunReq

2

calculate Speed ramp step size (Hz per scan)

MUL
Source A Ramp_Accel_HzPerSec
0.0
Source B Task_DT
0.0
Dest SpeedStep
0.0

3

Ramp speed up toward commad reference

Drive_RunReq

LE
Source A Conv_SpeedFb_Hz
0.0
Source B Ref_Hz
0.0

ADD
Source A Conv_SpeedFb_Hz
0.0
Source B SpeedStep
0.0
Dest Conv_SpeedFb_Hz
0.0

4

Ramp speed down when drive run request is removed

Drive_RunReq

SUB
Source A Conv_SpeedFb_Hz
0.0
Source B SpeedStep
0.0
Dest Conv_SpeedFb_Hz
0.0

5

Clamp speed to zero if deceleration overshoots

LE
Source A Conv_SpeedFb_Hz
0.0
Source B 0

MOVE
Source 0
Dest Conv_SpeedFb_Hz
0.0

6

Convert simulated speed (Hz) into scaled VFD feedback value

MUL
Source A Conv_SpeedFb_Hz
0.0
Source B 100
Dest SpeedStep
0.0

MOVE
Source SpeedStep
0.0
Dest VFD_Feedback
0

7

Conveyor considered running when speed exceeds minimum threshold

GE
Source A Conv_SpeedFb_Hz 0.0
Source B 0.2

Conv_RunFb

8

Calculate speed error between reference and feedback

SUB
Source A Ref_Hz 0.0
Source B Conv_SpeedFb_Hz 0.0
Dest Temp_SpeedError 0.0

ABS
Source Temp_SpeedError 0.0
Dest SpeedError 0

LE
Source A SpeedError 0
Source B 0.5

Conv_AtSpeed

9

set AtSpeed when speed error is within acceptable tolerance

MOVE
Source 0
Dest VFD_LogisticStatus 0

Sim_DriverFault / Sim_CommFault / VFD_LogisticStatus.0

Conv_RunFb — VFD_LogisticStatus.1 (L)

Drive_RunReq — VFD_LogisticStatus.2 (L)

Sim_DriverFault — VFD_LogisticStatus.7 (L)

Conv_AtSpeed — VFD_LogisticStatus.8 (L)

(End)